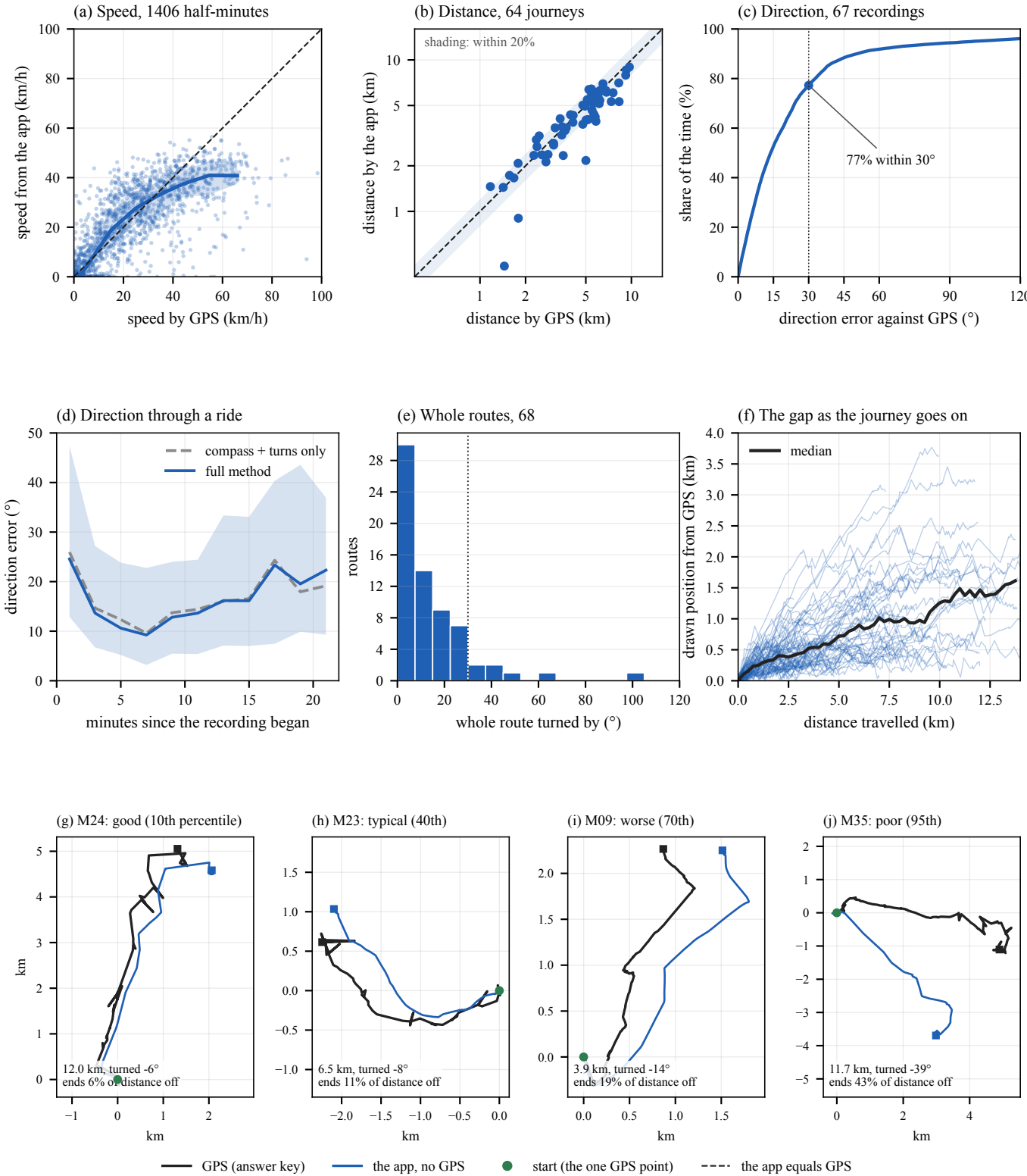

(a) Speed, 1406 half-minutes
speed from the app (km/h)
speed by GPS (km/h)
(b) Distance, 64 journeys
shading: within 20%
distance by the app (km)
distance by GPS (km)
(c) Direction, 67 recordings
share of the time (%)
77% within 30°
direction error against GPS (°)
(d) Direction through a ride
compass + turns only
full method
direction error (°)
minutes since the recording began
(e) Whole routes, 68
routes
whole route turned by (°)
(f) The gap as the journey goes on
median
drawn position from GPS (km)
distance travelled (km)
(g) M24: good (10th percentile)
12.0 km, turned -6°
ends 6% of distance off
(h) M23: typical (40th)
6.5 km, turned -8°
ends 11% of distance off
(i) M09: worse (70th)
3.9 km, turned -14°
ends 19% of distance off
(j) M35: poor (95th)
11.7 km, turned -39°
ends 43% of distance off
km
GPS (answer key)
the app, no GPS
start (the one GPS point)
the app equals GPS